

Xi Guo

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EDUCATION

University of Washington

Master of Science in Electrical and Computer Engineering

Seattle, WA

Sep 2026 — Jun 2028

Huazhong University of Science and Technology

Bachelor of Engineering in Computer Science and Technology

Wuhan, China

Sep 2019 — Jun 2023

PROFESSIONAL EXPERIENCE

PDD Holdings *A global e-commerce company (Nasdaq: PDD), parent of Pinduoduo and Temu.*

Shanghai, China

Software Engineer, Machine Learning Infrastructure

Jul 2023 — Nov 2025

- Worked on large-scale **training infrastructures** for e-commerce search and recommendation models serving **300M+ DAU**, with a focus on distributed training frameworks, sparse embedding systems, and training performance optimization.
- Collaborated closely with algorithm teams to optimize training workflows and diagnose production model training issues.

■ Large-scale Training for Generative Recommendation

- Built **PyTorch/TorchRec** training workflows for a generative recommender (HSTU), scaling billion-parameter sparse training across recall, CTR, and CVR tasks with DDP and TorchRec DMP
- Implemented pipelined training execution to overlap CPU data preprocessing, sparse embedding lookup, forward/backward computation, and parameter updates, improving training throughput by **40%**.
- Optimized sparse embedding communication by deduplicating feature IDs before TorchRec embedding lookup, improving 32 H100 GPU end-to-end throughput by **25%**.

■ Training Framework Migration (TensorFlow → PyTorch)

- Migrated a recommendation training pipeline from TensorFlow to PyTorch on **multi-node H100 clusters**, rebuilding data loading, sharded checkpointing, and cross-framework consistency checks.
- Integrated custom **C++ sparse Parameter Server** as a CPU backend to offload massive read-only embedding tables, avoiding GPU memory constraints for memory-intensive downstream models.
- Optimized **data prefetching** and CPU-GPU coordination, increasing GPU utilization by **30%+** and reducing training time by **15%** versus the legacy TensorFlow pipeline.

■ Semi-Synchronous Training Mechanism

- Extended TensorFlow **SyncReplicaOptimizer** with a semi-sync mechanism, implementing global batch aggregation, token-based gradient admission and staleness control while allowing workers to progress independently.
- Improved training throughput by **60%** over synchronous training on a production CVR model, reaching **85%** of asynchronous training throughput while observing a **0.3%** AUC improvement over the asynchronous baseline.
- Supported parameter-level update strategies by TensorFlow graph modifications to meet model-specific requirements.

■ Distributed Training Infrastructure

- Built multi-node training startup coordination with **ZooKeeper** and RPC, using CAS-based leader election to assign a primary worker for code distribution and cluster metadata collection.
- Coordinated worker readiness and generated TF_CONFIG for multi-node training jobs, enabling reliable cluster bring-up.
- Developed backend APIs(Python/Java) for training job submission, resource management, and access control, and implemented proxy routing to unify service endpoints.

SenseTime *An AI company (HKEX: 0020) specializing in computer vision and deep learning.*

Shanghai, China

Software Engineer Intern

Jul 2022 — Sep 2022

- Developed Golang (Gin) backend services and **REST APIs** for an enterprise deep learning platform, supporting algorithm management, training workflows, and access control.
- Built and maintained **Helm**-based deployment workflows for **Kubernetes** services and training jobs, including chart validation, environment configuration, and release deployment.
- Integrated pretrained models and training pipelines into the platform, implementing model configuration validation and supporting model artifact management with object storage.

SKILLS

Languages: C/C++, Python, Go, Java, SQL, Bash

Machine Learning: Deep Learning, Recommendation Systems, Sparse Embeddings, Transformers

ML Frameworks: PyTorch, TensorFlow, TorchRec, CUDA, Triton, NCCL, Distributed Training

Systems & Infra: Distributed Systems, Parameter Servers, Linux, Kubernetes, Docker, ZooKeeper, HDFS

Software & Tools: Git, MySQL, REST APIs, Microservices, Bazel